

NASIR SHARAF

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EXPERIENCE

Software Development Engineer at General Motors

Remote, Jun 2022 – Present

- Developed software as a contractor via Akkodis Staffing
- Using C++ and Python to develop in an embedded systems environment on driver assisted/autonomous vehicles as part of the connected vehicles research and development group
- Updating, maintaining, and refactoring high definition localizer module (HDLM) and developing features for crowdsourced map and map over the air (MOTA) applications for SuperCruise

Software Development Engineer at Theia Scientific

Remote, Oct 2021 – Mar 2022

- Provided automated workflows and machine learning solutions for laboratory and research scale microscopy
- Used Nvidia Jetson edge computing devices, Python, and Grafana to deploy PyTorch machine learning models for real time analysis
- Refactored existing python code base with Ray for scalable deployment of YOLOv5 algorithm

Graduate Research Assistant at Oakland University

Dept. of Computer Science and Engineering

Rochester, MI, Sep 2019 – May 2021

- Conducted cybersecurity research on malware in industrial controls systems
- Developed a physical ontology-based intrusion detection system (POBIDS) to capture malware targeting industrial hardware and published findings
- Developed an artificial immune system (ICSAIS) coupled with taint analysis to detect physics-based attacks in an electrical substation emulation

Associate Energy Engineer at Siemens Industry Inc

Plymouth, MI, Jul 2016 – Aug 2019

- Drafted and provided submittal drawings for Siemens Demand Flow Chiller Optimization and Siemens Green Labs projects. Largest single project valued at \$775,000
- Prepared submittals for systems level controls and bid documents for solutions construction projects as part of the Design Team, including parts sizing, specifications review, and custom drawing requests

PUBLICATIONS

K. G. Field, P. Patki, N. Sharaf, K. Sun, L. Hawkins, M. Lynch, R. Jacobs, D. D. Morgan, L. He, and C. R. Field, "Real-time, On-Microscope Automated Quantification of Features in Microscopy Experiments Using Machine Learning and Edge Computing," *Microscopy and Microanalysis*, vol. 28, no. S1, pp. 2046–2048, 2022.

[Thesis] N. Sharaf, "An Artificial Immune System Enhanced Taint Analysis for Industrial Control Systems", Oakland University, December 2021

M. Yahya, N. Sharaf, J. Rrushi, H. Tay, B. Liu, and K. Xu, "Physics Reasoning for Intrusion Detection in Industrial Networks", In *Proceedings of The Second IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications*, Virtual, December 2020

SKILLS

Software: Python • JavaScript • C • C++ • R • Matlab • Octave • Java • AutoCAD • Solidworks • Fusion360 • SQL • MongoDB • PostgreSQL

Hardware: 3D Printing • CNC Routing • Microcontrollers (Arduino, BeagleBone Black) • PLCS (Siemens S7) • Nvidia Jetson Nano • Nvidia Jetson Xavier

PROJECTS

[Rashomon](#) - A "newsbetting" site in which users bet on the political bias of the sources of de-titled news articles. The purpose is to prevent the spread of disinformation by designing a system that made users think about their biases. Developed in Django, PostgreSQL, and Bootstrap. Deployed on DigitalOcean

[Solarstack](#) – A patented design that consists of a wooden enclosure that held several solar panels, light guides, and linear cylindrical lenses that aims increase energy density and maximize solar power production

ICSAIS - An artificial immune system enhanced taint analysis to analyze an electrical substation emulation for possible attack surfaces for physics-based attacks. Subject of master's thesis

EDUCATION

**Oakland University,
Rochester, MI**

Master of Science in Computer Science, Aug 2021

**University of Pittsburgh,
Pittsburgh, PA**

Bachelor of Science in Mechanical Engineering, Dec 2015